

tions, price point, and functionality, allowing us now to gain traction with many good customers.

What solutions do you currently provide?

We have two products right now—the Sherpa Tug and Sherpa RollerTop. One moves trolleys autonomously inside factories, while the other moves plastic bins in factories.

Could you explain your technology to our engineering audiences?

To put that simply, an autonomous vehicle needs to ask three questions in a loop—where is it, what is around it, and then what will its next step be that will take it towards its goal? As long as you can answer these questions successfully in a tight loop, you will get to your location. That at a very high level is what we do. We need to ask these questions every 100 milliseconds and we do that. Our product is ISO 3691-4 certified. That is the best safety standard for such vehicles available. Safety is very important, especially in automotive companies

Why do we need this technology?

Generally, some material movement in factories is a repetitive, boring, hazardous, and backbreaking activity. The fundamental thing is that human beings are not supposed to do this repetitive activity which involves high fatigue. We cannot do it at the right accuracy level as people will get bored. Also, the number of people required to do these things with time is not increasing. This becomes the genesis of the demand for such things. The other reason is the volume of production. It is not possible to get the right volume and quality of production, without the correct automation. One cannot achieve this by just putting more people on the process. After some time, you have to automate many of these things to get the required throughput and volume. We are a part of the overall journey of

automation, which is a natural journey for any industry as it matures, scales up and gains volume. This becomes the core need for the kind of product that we are making.

We see bots acting as delivery agents in the US in universities. Will we ever see that in India and how soon can that be?

Frankly, bots doing delivery is very feasible. The challenge is not the technology but good footpaths. In India, unfortunately, we don't have good footpaths. When India gets good footpaths, this technology won't be too complicated to be ready for deployment. I would say the limiting part is not the bot technology but the footpath technology!

What is the need for such systems in developing countries such as India, where labour is easily available?

That is a good question. It is a misconception that labour is very easily available in India. You talk to any business and there's always a shortage of skilled manpower. Also, the Indian average ages look very good on paper. People will often tell you that the average age in India is just 26 years or 27 years, but the average age in Tamil Nadu is 35 years. The average age in Karnataka is also very similar now. These are the more industrialised areas of our country. That is one part. The other part is that youngsters who are coming out of college, and for a good reason, aspire for better jobs. Today's youngsters don't want to do a mind-numbing job of just moving material from one part to another where there is no application of the brain. This is not really very dignified work for human beings and over time people should not be doing this kind of work. So it's not a question of 'whether' people are available or not to do work. It's a question of 'should' people be doing this kind of work or not.

You're talking about skilled labour. What about unskilled labour? We do have plenty of that in India.

Yeah, that is true, but in a factory, you have to be skilled. You can't operate in a modern factory unless you have some skills. You have to understand discipline. When we say skilled and unskilled, it's not just knowing how to operate complex machinery and tools. It is accompanied by the discipline of working in a factory. If you have to do something every 15 minutes it cannot be 16 or 14. These kinds of work habits and discipline come as a process of becoming skilled. Unskilled people don't have these kinds of work delivery habits, which are mandatory things today for modern manufacturing. What we need to do is increase our ITI training output from 2.5 million per annum to 10 million per annum on a mission mode. The more skilled people we have in the country across areas, the wealthier we are as a nation.

Amazon and Walmart have started drone deliveries in certain parts of the US, which are supervised by skilled employees. Earlier deliveries were conducted by unskilled labour. In India, these transportation jobs are done by people whose livelihood depends on it. Don't you think this is eating into their job market?

For the US, the answer is even clearer. They just don't have enough people. For India, we had people doing the transportation job earlier. That's the evolution of the human journey. People would keep moving towards higher-value addition jobs, and we'd need to keep automating our lower-value addition jobs. The economy yields better gains by adding more productivity per person. The only magic humankind knows is technology. It has happened before as well—somebody invented the wheel; whatever we could do manually in a field, we could do better with a bullock cart, and even better with the